SALES AND CUSTOMER ANALYTICS SYSTEM

Project Title: ToyWorld – E-Commerce Management System

Objectives

- Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
- Generate total sales reports.
- Find top customers based on purchase amount.
- Identify best-selling toys.
- Perform category-wise sales analysis.

1. Aggregate Functions

Definition

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result. In the ToyWorld E-Commerce Management System, aggregate functions are used to analyze toy sales, customer orders, and product details.

1.1 COUNT()

Find the total number of orders placed by customers.

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;
```

```
TOTAL_ORDERS
------------
           8
```

1.2 SUM()

Find the total sales amount generated from toy orders.

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;
```

```
TOTAL_SALES
-----------
       8950
```

1.3 AVG()

Find the average order amount.

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;
```

```
AVERAGE_ORDER_AMOUNT
--------------------
             1118.75
```

1.4 MIN()

Find the lowest order amount.

```
SQL> SELECT MIN(Total_Amount) AS Lowest_Order_Amount
  2  FROM Orders;
```

```
LOWEST_ORDER_AMOUNT
-------------------
                650
```

1.5 MAX()

Find the highest order amount.

```
SQL> SELECT MAX(Total_Amount) AS Highest_Order_Amount
  2  FROM Orders;
```

```
HIGHEST_ORDER_AMOUNT
--------------------
                1800
```

2. Generate Total Sales Reports

A sales report can be generated using the SUM() function on the Total_Amount column.

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;
```

```
TOTAL_SALES
-----------
       8950
```

This query calculates the total sales amount generated by ToyWorld through customer orders.

2.1 Customer-wise Sales Report

To find how much each customer has purchased:

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.ID = O.CUSTOMER_ID
  8  GROUP BY C.ID, C.NAME
  9  ORDER BY TOTAL_PURCHASE DESC;
```

```
     ID NAME                           TOTAL_PURCHASE
------- ------------------------------ --------------
      2 Khilan                                   2650
      1 Ramesh                                   2450
      3 kaushik                                  2250
      5 Hardik                                    950
      4 Chaitali                                  650
```

Purpose

This report shows:

- Customer ID
- Customer name
- Total purchase amount
- Customers arranged according to purchase amount

3. Find Top Customers Based on Purchase Amount

The SUM() function can be combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts.

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.ID = O.CUSTOMER_ID
  8  GROUP BY C.ID, C.NAME
  9  ORDER BY TOTAL_PURCHASE DESC
 10  FETCH FIRST 5 ROWS ONLY;
```

```
     ID NAME                           TOTAL_PURCHASE
------- ------------------------------ --------------
      2 Khilan                                   2650
      1 Ramesh                                   2450
      3 kaushik                                  2250
      5 Hardik                                    950
      4 Chaitali                                  650
```

The customer appearing at the top has the highest purchase amount in the generated report.

4. Identify Best-Selling Toys

The Order_Details table contains the quantity of each toy ordered.

Therefore, we can use SUM(Quantity) to find the total quantity sold for each toy.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDER_ITEM OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_QUANTITY_SOLD DESC;
```

```
PRODUCT_ID PRODUCT_NAME                                                                                          TOTAL_QUANTITY_SOLD
---------- ------------------------------------------------------------------------------------------------------ -------------------
       101 ABC Learning Kit                                                                                                         5
       103 Remote Car                                                                                                               5
       109 Superhero Figure                                                                                                         2
       105 Football                                                                                                                 1
       104 Chess Board                                                                                                              1
```

Purpose

This query:

1. Connects Product and Order_Details.
2. Groups records according to toy products.
3. Calculates total quantity sold.
4. Displays toys from highest to lowest quantity sold.

4.1 Product Sales Amount

Instead of only calculating quantity, we can calculate the total sales generated by each toy product.

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
  5       SUM(OD.QUANTITY * OD.UNIT_PRICE) AS TOTAL_SALES
  6  FROM PRODUCT P
  7  JOIN ORDER_ITEM OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
 10  ORDER BY TOTAL_SALES DESC;
```

```
PRODUCT_ID PRODUCT_NAME                                                                                          TOTAL_QUANTITY TOTAL_SALES
---------- ------------------------------------------------------------------------------------------------------ -------------- -----------
       103 Remote Car                                                                                                          5        4150
       101 ABC Learning Kit                                                                                                    5        2400
       104 Chess Board                                                                                                         1        1500
       109 Superhero Figure                                                                                                    2        1100
       105 Football                                                                                                            1         650
```

This gives both quantity sold and sales amount for each toy.

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each toy category.

```
SQL> SELECT
  2       C.CATEGORY_ID,
  3       C.CATEGORY_NAME,
  4       SUM(OD.QUANTITY * OD.UNIT_PRICE) AS CATEGORY_SALES
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  JOIN ORDER_ITEM OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 11  ORDER BY CATEGORY_SALES DESC;
```

```
CATEGORY_ID CATEGORY_NAME                                          CATEGORY_SALES
----------- ---------------------------------------------------- --------------
          3 Remote Control Toys                                            4150
          1 Educational Toys                                               2400
          4 Board Games                                                    1500
          9 Action Figures                                                 1100
          5 Outdoor Toys                                                    650
```

This query helps identify which toy category generates the highest sales in the ToyWorld E-Commerce Management System.

6. Combined Business Sales Report

A complete sales report can combine customer, order, product, and category information.

```
SQL> SELECT
  2      C.NAME AS CUSTOMER_NAME,
  3      O.ORDER_ID,
  4      P.PRODUCT_NAME,
  5      CAT.CATEGORY_NAME,
  6      OD.QUANTITY,
  7      OD.UNIT_PRICE,
  8      (OD.QUANTITY * OD.UNIT_PRICE) AS TOTAL_AMOUNT
  9  FROM CUSTOMER C
 10  JOIN ORDERS O
 11  ON C.ID = O.CUSTOMER_ID
 12  JOIN ORDER_ITEM OD
 13  ON O.ORDER_ID = OD.ORDER_ID
 14  JOIN PRODUCT P
 15  ON OD.PRODUCT_ID = P.PRODUCT_ID
 16  JOIN CATEGORY CAT
 17  ON P.CATEGORY_ID = CAT.CATEGORY_ID;
```

```
CUSTOMER_NAME                  ORDER_ID PRODUCT_NAME
------------------------------ ---------- ------------------------------------
Ramesh                               1001 ABC Learning Kit
Ramesh                               1006 Superhero Figure
Khilan                               1002 Remote Car
Khilan                               1007 ABC Learning Kit
kaushik                              1003 Chess Board
kaushik                              1008 Remote Car
Chaitali                             1004 Football

7 rows selected.
```

```
              CATEGORY_NAME    QUANTITY UNIT_PRICE TOTAL_AMOUNT
--------------- ---------------------------------------------- ---------- ---------- ------------
              Educational Toys                                          2        300          600
              Action Figures                                            2        550         1100
              Remote Control Toys                                       4        850         3400
              Educational Toys                                          3        600         1800
              Board Games    1        1500         1500
              Remote Control Toys                                       1        750          750
              Outdoor Toys   1         650          650
```

Report Structure

This report can be used to understand:

- Who purchased the toy
- Which order was placed
- Which toy product was purchased
- Which category the toy belongs to
- Quantity purchased
- Amount generated

Conclusion

The ToyWorld E-Commerce Management System helps manage customer orders, toy products, and sales details efficiently using SQL aggregate functions and queries.